Zhihao Chen

MSc Data Science Student and AI Researcher

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Education

MSc — **Federal Institute of Technology Lausanne (EPFL)**, Data Science
Lausanne, Switzerland
Sept 2026 - present

BSc — **Polytechnic University of Catalonia (UPC)**, Artificial Intelligence
Barcelona, Spain
Sept 2022 - June 2026
- GPA: 9.00/10.0 (Top 10%)

Exchange — **Federal Institute of Technology Lausanne (EPFL)**, Computer Science
Lausanne, Switzerland
Sept 2025 - Feb 2026
- GPA: 5.93/6.0

Experience

VILAB - EPFL, Graduate Student Researcher
Lausanne, Switzerland
Sept 2026 - present
1 month
Continuous Generative Modeling for Unified Any-to-Any Multimodal Systems
- Extending unified any-to-any multimodal architectures via continuous generative processes and flow-based decoders across heterogeneous sensory modalities.

EPFL, Student Teaching Assistant
Lausanne, Switzerland
Sept 2026 - present
1 month
EE-556 Mathematics of Data: From Theory to Computation

LAMP - CVC - UAB, Undergraduate Student Researcher
Barcelona, Spain
Feb 2026 - Sept 2026
8 months
Unified Multi-Modal Continual Learning
- Formulated a sequential benchmark to measure cross-modal catastrophic forgetting between visual reasoning and image synthesis in unified multimodal generative models.
- Found that decoupled architectures mitigate bidirectional representation degradation compared with monolithic architectures.
- Awarded the Elena Maseras Research Scholarship 2026 for academic excellence (13% acceptance rate).

IRI - UPC, Undergraduate Student Researcher
Barcelona, Spain
Feb 2026 - Sept 2026
8 months
Future Action Prediction in Video Models
- Developed WarmFlow, an encoder-initialized flow-matching framework that replaces pure-noise diffusion with dynamic structural priors for action segmentation.
- Achieved competitive generative coherence with one function evaluation (NFE=1), reducing computation by 10x-25x for latency-critical applications.

BioRob - EPFL, Visiting Student Researcher
Lausanne, Switzerland
Sept 2025 - Feb 2026
6 months
Context-Aware Spatial Action Recognition
- Extended the SkateFormer architecture for multimodal spatial action recognition, using geometric priors to generalize to occluded layouts.
- Curated a 10k-sample dataset and deployed the framework on a ROS2 mobile swarm, achieving a 20% accuracy gain over baselines and 25ms latency for closed-loop control.

PayRetailers, AI Engineer Intern
Barcelona, Spain
June 2025 - Aug 2025
3 months
Multi-Agent Systems and LLM Analysis
- Evaluated prompting strategies, including chain-of-thought and few-shot prompting, and connected LLMs to external tools through multi-agent workflows.
- Built internal B2B AI solutions, including retrieval-augmented generation systems and automated analytical pipelines.

Publications

Continual Adaptation Across Modalities: Learning and Forgetting in Unified Multimodal Models
Xinrui Wang, **Zhihao Chen**, Jia Tang, Alexandra Gomez-Villa, Bartłomiej Twardowski, Joost van de Weijer, Songcan Chen
(NeurIPS 2026 Workshop on CL4FMAgents)

Projects

B2Travel - HackUPC 2026
Built an LLM-grounded multimodal recommendation system that maps CLIP visual-semantic embeddings into a 3D topological space for spatial retrieval.
- Second Prize, Skyscanner Challenge; Best Use of ElevenLabs.

MXEgg - LauzHack 2025
Built an LLM-driven multi-agent simulation with a custom physics engine and localized vision to study emergent cooperative behavior.
- Overall finalist, LauzHack 2025.

AIvidence - MERIThon 2025 AI Challenge
Built a retrieval-grounded multi-agent framework that distributes chain-of-thought reasoning across specialized LLMs for deepfake detection.
- First Prize, MERIThon 2025 AI Challenge.

Awards

Elena Maseras Research Scholarship 2026 — LAMP - CVC - UAB
Awarded by LAMP, CVC-UAB for academic excellence (13% acceptance rate).

Excellence Scholarship — Santander 360º

Second Prize, Skyscanner Challenge and Best Use of ElevenLabs — HackUPC 2026

Overall Finalist, LauzHack 2025 — EPFL, Switzerland
MXEgg: Built an LLM-driven multi-agent simulation with a custom physics engine and localized vision to study emergent cooperative behavior.

First Prize, MERIThon 2025 AI Challenge — UPC, Spain
AIvidence: Built a retrieval-grounded multi-agent framework for deepfake detection.

Academic Honors — UPC, Spain
Four courses with distinction.

Skills

Programming: Python, C, C++, MATLAB, SQL, R

Machine Learning and Data: PyTorch, Scikit-learn, Hugging Face, OpenCV, LangChain, Weights & Biases, DDP/FSDP

Systems and Engineering: Jupyter, LaTeX, Git/GitHub, Docker, SLURM

Languages

Chinese: Native proficiency

Spanish: Native proficiency

Catalan: Native proficiency

English: Professional proficiency (C1)

French: Elementary proficiency (A2)